

Basic of EM waves

Question 1: Describe the Maxwell equations

Answer 1: $\nabla \times \bar{e} = -\frac{\partial \bar{b}}{\partial t}$

$$\nabla \cdot \bar{d} = \rho$$

$$\nabla \times \bar{h} = \bar{j} + \frac{\partial \bar{d}}{\partial t}$$

$$\nabla \cdot \bar{b} = 0$$

- $\bar{e}(\bar{r}, t)$ is ELECTRIC FIELD $[V/m]$
- $\bar{d}(\bar{r}, t)$ is the electric flux density $[C/m^2]$
- $\bar{h}(\bar{r}, t)$ is MAGNETIC FIELD $[A/m]$
- $\bar{b}(\bar{r}, t)$ is MAGNETIC FLUX DENSITY $[T = Wb/m^2]$
- $\rho(\bar{r}, t)$ is CHARGE DENSITY $[C/m^3]$
- $\bar{J}(\bar{r}, t)$ is ELECTRIC CURRENT DENSITY $[A/m^2]$
- $\bar{r} = (x, y, z)$

Assumptions: $\rho = 0, \bar{J} = 0$

Question 2: EM fields in harmonic domain

Answer 2: In many cases we can limit the study of EM fields to the harmonic signals. It is convenient to represent fields by their phaser using the following relation:

$$\bar{e}(\bar{r}, t) = \text{Re} [\bar{E}(\bar{r})e^{j\omega t}] \quad (1)$$

In this case we have no electric charges or current. Hence:

$$\nabla \times \bar{E} = -j\omega \bar{B} = -j\omega \mu \bar{H}$$

$$\nabla \cdot \bar{D} = 0$$

$$\nabla \times \bar{H} = j\omega \bar{D} = j\omega \epsilon \bar{E}$$

$$\nabla \cdot \bar{B} = 0$$

Question 3: Describe the constitutive relations

Answer 3: The relation between fields and flux densities depend on the medium where the fields propagate. In general we have:

$$\bar{D} = \epsilon \bar{E}$$

$$\bar{B} = \mu \bar{H}$$

where ϵ is the electric permittivity and μ is the magnetic permeability of the medium. $[F/m]$ and $[H/m]$ respectively.

Question 4: Describe Helmholtz equation

Answer 4: The Helmholtz equation is a second-order partial differential equation that describes the propagation of waves in a medium. Starting from the Maxwell equations in harmonic domain without charges and currents, we can derive the Helmholtz equation for the electric and magnetic fields.

Maxwell's Equations (Phasor Form)

$$\nabla \times \bar{E} = -j\omega\mu\bar{H} \quad \nabla \cdot \bar{E} = 0$$

$$\nabla \times \bar{H} = j\omega\epsilon\bar{E} \quad \nabla \cdot \bar{H} = 0$$

Derivation of the Helmholtz Equation

1) Apply the curl to Faraday's Law:

$$\nabla \times (\nabla \times \bar{E}) = \nabla \times (-j\omega\mu\bar{H})$$

2) Due to the linearity of the operator, move the constants outside:

$$\nabla \times (\nabla \times \bar{E}) = -j\omega\mu(\nabla \times \bar{H})$$

3) Substitute the curl of $\bar{H}$ using the Ampère-Maxwell Law:

$$\nabla \times (\nabla \times \bar{E}) = -j\omega\mu(j\omega\epsilon\bar{E})$$

4) Simplify the constant terms (given that $j^2 = -1$):

$$\nabla \times (\nabla \times \bar{E}) = \omega^2\mu\epsilon\bar{E}$$

5) Use the vector identity $\nabla \times \nabla \times \bar{A} = \nabla(\nabla \cdot \bar{A}) - \nabla^2\bar{A}$:

$$\nabla(\nabla \cdot \bar{E}) - \nabla^2\bar{E} = \omega^2\mu\epsilon\bar{E}$$

Since $\nabla \cdot \bar{E} = 0$ in a source-free medium, the gradient term vanishes:

$$-\nabla^2\bar{E} = \omega^2\mu\epsilon\bar{E}$$

6) Rearrange the terms to obtain the Helmholtz Equation:

$$\nabla^2\bar{E} + \omega^2\mu\epsilon\bar{E} = 0$$

For magnetic field the steps are the same.

Question 5: What is the advantage of using piecewise homogeneous media?

Answer 5: Piecewise homogeneous is a simplification that allows us to consider ϵ and μ as constants in a given region. But normally ϵ and μ are not constant, so we solve this problem by using boundary conditions.

Question 6: What are the boundary conditions?

Answer 6: We consider the thin surface between 2 different materials. We define the normal vector $\hat{n}$.

For flux densities

- **Density of magnetic flux:** normal component is always continuous

$$\hat{n} \cdot (\bar{B}_2 - \bar{B}_1) = 0 \quad (2)$$

- **Density of electric flux:** normal component is discontinuous

$$\hat{n} \cdot (\bar{D}_2 - \bar{D}_1) = \rho_s \quad (3)$$

For fields

- **Electric field:** tangential component is always continuous

$$\hat{n} \times (\bar{E}_2 - \bar{E}_1) = 0 \quad (4)$$

- **Magnetic field:** tangential component is discontinuous

$$\hat{n} \times (\bar{H}_2 - \bar{H}_1) = \bar{J}_s \quad (5)$$

For our assumptions, $\rho_s = 0$ and $\bar{J}_s = 0$.

Question 7: Describe the Poynting vector

Answer 7: In the time domain the Poynting vector is defined as:

$$\bar{P}(\bar{r}, t) = \bar{E}(\bar{r}, t) \times \bar{H}(\bar{r}, t) \quad (6)$$

where $\bar{E}$ is the electric field and $\bar{H}$ is the magnetic field.

While in the harmonic domain we have:

$$\bar{P}(\bar{r}) = \frac{1}{2} \bar{E}(\bar{r}) \times \bar{H}^*(\bar{r}) \quad (7)$$

where $\bar{H}^*$ is the complex conjugate of $\bar{H}$.

In other words, it represents the power flow per unit area.

Question 8: Describe the polarization of EM waves and the polarization plane

Answer 8: A generic sinusoidal EM field can be written as:

$$\vec{e}(\vec{r}, t) = \sum_{n=1}^{3 \text{ dim}} C_n(\vec{r}) \cos(\omega t + \phi_n(\vec{r})) \hat{x}_n = \underbrace{c'(\vec{r}, t)}_{\vec{E}_{\text{Re}}} \cos(\omega t) - \underbrace{c''(\vec{r})}_{\vec{E}_{\text{Im}}} \sin(\omega t) \quad (8)$$

which clearly shows that the electric vector of a generic sinusoidal field lays at any time on a plane defined by the vectors $\vec{E}_{\text{Re}}$ and $\vec{E}_{\text{Im}}$.

This is known as polarization plane: constant in time, vary in space. The sinusoidal field is periodic, so the curve it describes must be closed. Generally, it is an ellipse.

This ellipse is described by two components:

$$\vec{e}'(t) = \begin{bmatrix} a \cos(\omega t + \phi) \\ b \sin(\omega t + \phi) \end{bmatrix} \quad (9)$$

We that the polarization is:

- Right handed: when the field rotates counterclockwise
- Left handed: when the field rotates clockwise

Two notable cases are:

- Linear polarization: when the ellipse degenerates into a line
 - Circular polarization: when the ellipse degenerates into a circle ($a = \pm b$)
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Question 9: Describe Jones vector and their utility

Answer 9: It's a bidimensional complex vector which describes a generic sinusoidal EF when represented with respect to its polarization plane. The Jones vector reads:

$$\vec{J} = \begin{bmatrix} a \cos(\delta) + jb \sin(\delta) \\ a \sin(\delta) - jb \cos(\delta) \end{bmatrix} \quad (10)$$

- Φ does not influence the polarization, but it's useful to describe field propagation;
 - Two polarizations are orthogonal if the corresponding Jones vectors are orthogonal. ($\vec{E}_1 \cdot \vec{E}_2 = 0$)
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Question 10: Describe Stokes vector

Answer 10: Stokes vectors are real vectors which describes a more intuitive geometrical representation of light polarization. The Stokes vector reads:

$$\vec{S} = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} |E_1|^2 + |E_2|^2 \\ |E_1|^2 - |E_2|^2 \\ 2\text{Re}[E_1 E_2^*] \\ 2\text{Im}[E_1 E_2^*] \end{bmatrix} = \begin{bmatrix} a^2 + b^2 \\ (a^2 - b^2) \cos 2\delta \\ (a^2 - b^2) \sin 2\delta \\ 2ab \end{bmatrix} \quad (11)$$

- S_0 total intensity;
- S_1 difference in intensities between horizontal and vertical linearly polarized components;

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- S_2 difference in intensities between linearly polarized components oriented at $+45^\circ$ and -45° ;
 - S_3 difference in intensities between right and left circularly polarized components

It does not depend on phase.

$$S_0 = \sqrt{S_1^2 + S_2^2 + S_3^2} \quad (12)$$

Monochromatic case: ω is constant.

Also, knowing the redundancy of S_0 , we can define the unit Stokes vector:

$$\hat{S} = \begin{bmatrix} S_1 \\ S_2 \\ S_3 \end{bmatrix} = \frac{1}{S_0} \begin{bmatrix} S_1 \\ S_2 \\ S_3 \end{bmatrix} = \frac{1}{a^2 + b^2} \begin{bmatrix} (a^2 - b^2) \cos 2\delta \\ (a^2 - b^2) \sin 2\delta \\ 2ab \end{bmatrix} \quad (13)$$

which can be further simplified with ellipticity: $\epsilon = \tan^{-1}(b/a)$

$$\tan \epsilon = \frac{b}{a} \quad (14)$$

$$\cos 2\epsilon = \frac{1 - \tan^2 \epsilon}{1 + \tan^2 \epsilon} = \frac{1 - (b/a)^2}{1 + (b/a)^2} = \frac{a^2 - b^2}{a^2 + b^2} \quad (15)$$

$$\sin 2\epsilon = \frac{2 \tan \epsilon}{1 + \tan^2 \epsilon} = \frac{2(b/a)}{1 + (b/a)^2} = \frac{2ab}{a^2 + b^2} \quad (16)$$

$$\hat{s} = \frac{1}{a^2 + b^2} \begin{bmatrix} (a^2 - b^2) \cos 2\delta \\ (a^2 - b^2) \sin 2\delta \\ 2ab \end{bmatrix} \quad (17)$$

$$\hat{s} = \begin{bmatrix} \cos 2\epsilon \cos 2\delta \\ \cos 2\epsilon \sin 2\delta \\ \sin 2\epsilon \end{bmatrix} \quad (18)$$

Question 11: Describe the Poincaré sphere

Answer 11: The unit stokes vector can be geometrically visualized as lying on the surface of a sphere of unit radius. Each point on the sphere represents a different polarization state.

$$\hat{S} = \begin{bmatrix} \cos 2\epsilon \cos 2\delta \\ \cos 2\epsilon \sin 2\delta \\ \sin 2\epsilon \end{bmatrix} \quad (19)$$

the various polarizations are:

- Equator: linear polarization
 - North pole: circular polarization right handed
 - South pole: circular polarization left handed
 - North hemisphere: elliptical polarization right handed
 - South hemisphere: elliptical polarization left handed
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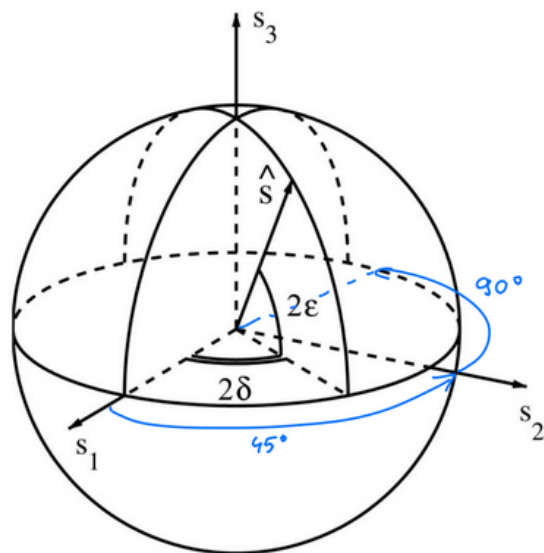


Figure 1: Poincaré sphere

Modal theory

Question 12: What is Total Internal Reflection?

Answer 12: Consider a fiber. We have the core and the cladding. We have to respect two conditions:

- The refractive index of the incidence medium must be greater than the refractive index of the transmission medium ($n_1 > n_2$);
- The angle of incidence θ_i must be greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

When $\theta_i < \theta_c$ we have *uniform trasmitted waves* and $n_1 \sin(\theta_i) = n_2 \sin(\theta_t)$.

Question 13: Considered a step-index fiber; let us determine the maximum input angle that a ray may have so that the ray refracted in the fiber core is guided by TIR.

Answer 13:

- **Snell's Law at the input interface (Air to Core):**

$$n_0 \sin \beta_M = n_1 \sin \beta_l \implies \sin \beta_M = n_1 \sqrt{1 - \cos^2 \beta_l} \quad (\text{assuming } n_0 \approx 1) \quad (20)$$

- **Total Internal Reflection (TIR) condition at Core-Cladding interface:** The ray is guided if the incidence angle is greater than or equal to the critical angle θ_c :

$$\frac{\pi}{2} - \beta_l \geq \theta_c \implies \cos \beta_l \geq \sin \theta_c \quad (21)$$

- **Critical Angle definition:**

$$\sin \theta_c = \frac{n_2}{n_1} \quad (22)$$

- **Substitution for Maximum Angle (β_M):** Substituting the limiting case $\cos \beta_l = \frac{n_2}{n_1}$ into the first equation:

$$\sin \beta_M = n_1 \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2} = \sqrt{n_1^2 - n_2^2} \quad (23)$$

- **Final Result (Numerical Aperture):**

$$NA = \sin \beta_M = \sqrt{n_1^2 - n_2^2} \quad (24)$$

Large NA $\implies$ easier to couple light into the fiber, but can cause ISI (modal dispersion).

Question 14: From S

Answer 14:
